

Heredity

➤ Heredity : →
The transmission of characters/traits from one generation to next generation is called HEREDITY.

* Variations : →
The difference in the characters (or traits) among the individual of a species is called Variation.

• Importance of Variation : →

- ① It increases the chance of survival in the changing environment.
- ② It is a basis of evolution.
- ③ It brings adaptation in organisms.

Note : → free ear lobes and attached ear lobes are two variants found in human population.

* Inherited Traits and Acquired Traits : →

Inherited Traits (germinal)	(somatic) Acquired Traits
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• These traits can be inherited and transferred to next generation.

• These traits are not inherited and cannot be transferred to next generation.

- These traits are inherited from parents to offspring.
- They are caused by a change in genes.
- for example: →
skin colour, eye colour, ear lobe.
- These traits are acquired after birth.
- They are not caused by the change in genes.
- for example: →
cut tail of mice, swimming skill, driving skill.

● **Genetics** : → It is a branch of biology that deals with the transmission of characters and variations is known as **GENETICS**.

father of Genetics : → **GREGOR MENDEL**.

* **DNA** (deoxyribonucleic acid) : → DNA is that molecule which carries genetic information for the development and functioning of an organism.

* **Chromosomes** : → Chromosomes is a thread-like structure present in the nucleus of a cell.

→ Total 23 pairs of chromosomes are present in humans.

- 22 pairs of chromosomes are identical in both males and females, they are called autosomes.
- 23rd pair of chromosomes is different in both males and females, it is called sex chromosome.
- 23rd pair in Male → XY
- 23rd pair in female → XX.
- Genes: → genes are unit of heredity which transfer characteristics from parents to offspring.
- Allele: → Gene which code for pair of contrasting traits.
for example: → Tall — TT
Dwarf — Tt
- * Dominant Gene: → The gene which decides the appearance of an organism. Even in the presence of a recessive gene.
- * Recessive Gene: → The gene which can decide the appearance of an organism only in the presence of another identical gene.

note: → TT → Tall tt → Dwarf.
Tt → Tall

* Genotype : →

Genotype is the description of genes present in an organism.

* phenotype : → The characteristics which is visible in an organism is called

phenotype

* Homozygous condition : →

The organism in which both gene are similar.

for example : →

TT : — Tall

tt : — dwarf

* Heterozygous condition : →

The organism in which both genes are different.

for example : →

Tt : — tall

note : →

- Dominant gene express itself in both homozygous and heterozygous condition.
- Recessive gene express itself only in the homozygous condition.

* Mendel and his work on Inheritance: Joban

- Gregor Mendel started his on plant breeding and hybridisation.
- He gave three laws of inheritance.
- Mendel is called the "father of genetics".
- Plant selected by Mendel : → Pea plant (*Pisum sativum*)

* Traits studied by Mendel : →

Character	dominant trait	Recessive trait
1. Seed shape.	Round	Wrinkled
2. Seed colour.	Yellow	Green
3. Flower colour.	Violet	White
4. Pod shape.	Inflated	Constricted
5. Pod colour.	Green	Yellow
6. Flower position.	Axial	Terminal
7. Stem length.	Tall	Dwarf.

questⁿ : → Why Mendel choose pea plant in his ~~expression~~ experiments?

- Pea plant has short life span so that it mature quickly.
- Pea plant has bisexual flower which is good to conduct self pollination or cross pollination.
- Pea plant has contrasting traits of seven characters.
- Large number of offspring can be produced.

~~Imp~~ ★ Monohybrid Cross : →

Cross b/w two pea plants with one pair of contrasting characters is called Monohybrid cross.

Parent's phenotype

Tall plant

Dwarf plant

gamete formation →

crossing of gametes →
during fertilisation

F₁ generation phenotype

F₁ generation cross

		T	t
♀	T	TT	Tt
	t	Tt	tt

F₂ generation phenotypic ratio →

Tall : dwarf
3 : 1

F₂ generation genotypic ratio →

TT : Tt : tt
1 : 2 : 1

- Observation of Monohybrid cross: $\rightarrow$
- All F_1 progeny were tall, no medium height plant.
- F_2 progeny $\frac{1}{4}$ ~~were short~~ were short $\frac{3}{4}$ were tall.
- phenotypic ratio F_2 generation 3:1
(3 tall, 1 short)
- Genotypic ratio F_2 generation 1:2:1
(TT, Tt, tt)

* Conclusion: $\rightarrow$

- TT and Tt are tall plants while tt is a short plant.
- Dominant trait express itself in both conditions homozygous and heterozygous condition while recessive trait express itself in only homozygous condition.

Dihybrid Cross: $\rightarrow$

It is a cross in which two characteristics is studied at time.

* Mendel studied two characteristics in dihybrid cross: →

1. seed color — (dominant) yellow (YY) (Recessive) Green (yy)

2. seed shape — Round (RR) wrinkled (rr)

[Parental Cross]

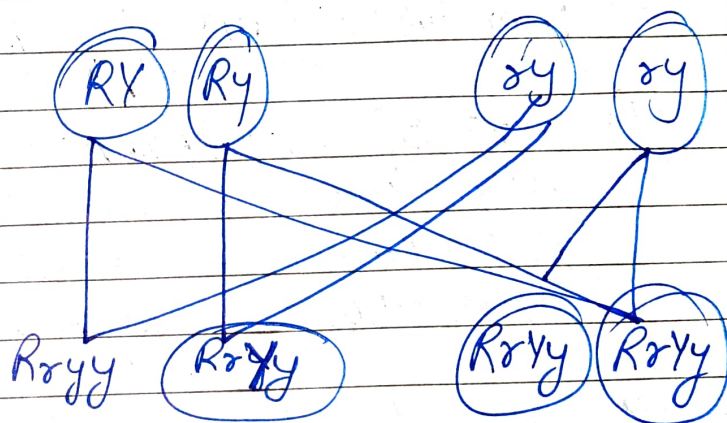
Parent's phenotype
allele
gamete.

Round
yellow
seed
 $R \cdot R Y Y$

wrinkled
green
seed
 $r r y y$

fusion of gamete →

F₁ generation
Phenotype



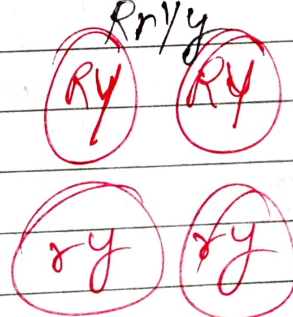
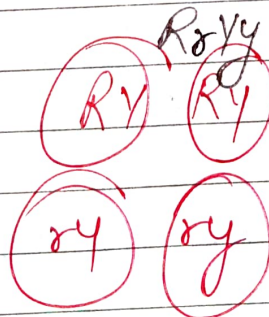
F₁ Cross

F₁ generation
phenotype

Round yellow
seed
 $R r Y y$

Round yellow
seed
 $R r Y y$

Gamete.



Generation

♀ $\rightarrow$	Ry	Ry	ry	ry
RY	RRYY	RRYy	RrYY	RrYy
Ry	RRYy	RRyy	RrYy	Rryy
ry	RrYY	RrYy	rrYY	rrYy
ry	RrYy	Rryy	rrYy	rryy

* Mendel's Law of Inheritance $\rightarrow$

(1) Law of Dominance $\rightarrow$

When an inherited pair of two alleles, the allele that express is called dominant and that does not express called recessive.

$\rightarrow$ It is based on monohybrid cross.

(2) Law of Segregation / Law of purity of gametes $\rightarrow$

During the gamete formation, copies of genes or alleles are divided such that each gamete receive only one allele.

$\rightarrow$ It is also based on monohybrid cross.

(3) Law of Independent Assortment :-
Alleles of two or more different genes get assorted into gametes independently of one another.

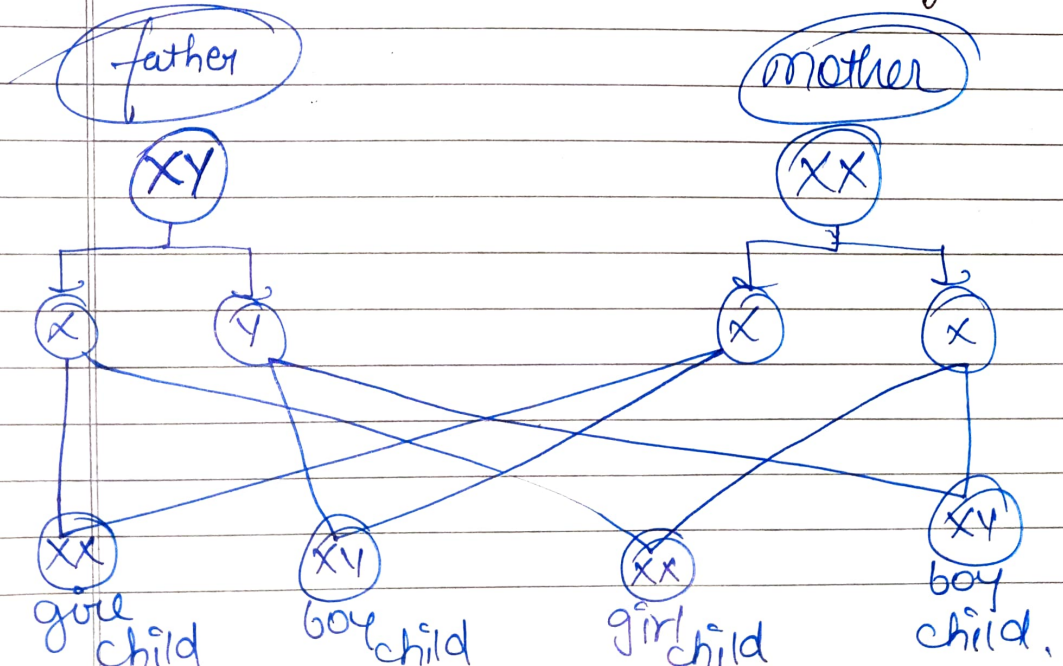
→ It is based on dihybrid cross.

• Sex Determination :-
The process by which the sex of a person is determined is called the sex determination.

* Environmental conditions :-
In turtle, alligators, crocodile, lizard gender is determined by temperature, at which fertilised eggs are kept.

→ In snails, individual can change sex.

* sex determination in human beings :-



OR

♀	X	XX	XX
♂	Y	XY	XY

• phenotypic ratio $\rightarrow$ $\frac{XX}{\text{male}} : \frac{XY}{\text{female}}$

• Genotypic ratio $\rightarrow$ $\frac{XX}{1} : \frac{XY}{1}$

* Blood group inheritance in human beings: $\rightarrow$

Phenotype / blood group

genotype

A

$I^A I^A$ or $I^A i$

B

$I^B I^B$ or $I^B i$

AB

$I^A I^B$

O

$i i$

* Intext questions pg — 143.

(1) Trait B have arisen earlier because its population is high as compare to the population of trait A. Trait B replicates itself earlier than Trait A.

(2) done ✓

pg → 147.

(1) (i) Law of Dominance
(ii) Explain monohybrid cross.

(2) (i) Law of Independent -
(ii) Explain dihybrid.

(3) Yes, These information is enough to tell blood group A is dominant and blood group O is recessive because it express itself only in homozygous condition.

Man (A)

Woman (O)

$I^A I^A$ or $I^A i$

ii

ii
 ii

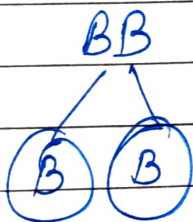
daughter (O)

Exercise questions →

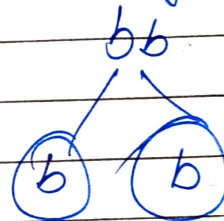
① (c) TtWw —

② These information is not enough to tell that light coloured eye is dominant. Recessive until the cross is made b/w light coloured eye and dark coloured eye.

③ male dog with black colour coat



female dog with white colour coat



	B	B
b	Bb	Bb
b	Bb	Bb

F₂ generation

→ All progeny are with black colour coat.

F₂ cross

	B	b
B	BB	Bb
b	Bb	bb

→ F₂ generation

white colour coat

• Phenotypic ratio Black 3 white
 3 1

• Genotypic ratio BB : Bb : bb
 1 2 1

④ Equal contribution of male & female is ensured in the progeny during sexual reproduction during the gamete formation the pair of allele divide and fuses equally with the other DNA.